Course Overview

CS3263

Foundations of Artificial Intelligence

Sem 2, AY2025-26

Syllabus

Responsible AI

Week	Topics	Week	Topics
1	Overview and Introduction - Responsible AI, Generative AI, Agentic AI - Human Factors in AI	7	Introduction to Generative Models - Generative Models, Foundation Models - Planning with Foundation Models
2	Knowledge Representation - Logic, Production Systems, Semantic Networks.	8	Mid-Term Test [12 March 2026]
3 Q	Planning - Classical Planning, Planning as Search and Inference	9	Decision Making over Time - Markov Decision Processes
4	Representing Uncertainty - Bayesian Networks, Conditional Independence	10	Decision Making with Learning - Reinforcement Learning
5	Reasoning under Uncertainty - Probabilistic Inference, Causal Modeling	11	Decision Making with Limited Resources (Online: NUS WB Day) - Constraint Satisfaction Problems, Online Algorithms
6 Q	Rational Decision Making - Decision Analysis, Value of Information	12 Q*	State of the Art - Human-AI Alignment, Future of AI and AI for Future
R		13	Project Presentations

Q: Quiz week Q*: In-class Quiz

Teaching Team

Name	Role
LEONG Tze Yun	Lecturer
VO Thanh Vinh	Co-Lecturer
FU Di	Tutor

Note: Please ask technical and course organization questions through **Discussions on CANVAS!**

Teaching Team – In the New AI Era

- Effective and responsible use of:
  - Generative AI Foundational Models
  - Search Engines
- Examples:
  - LLAMA
  - ChatGPT
  - Qwen
  - DeepSeek
  - …

…

Note: Please attribute References or Sources from Generative AI Models or Search Engines!

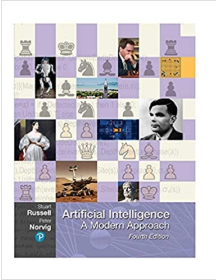
Course Logistics

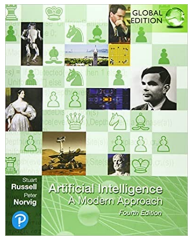
- Classes (weekly)
  - Thursday 1000 - 1200 LT15 Lecture
  - Mondays and Tuesdays T3, T4, T5, T6 Tutorial
- Grading policy
  - Homework, participation 20 + 10%
  - Quizzes (3 – *last one in-class*) 20%
  - Test [1000 to 1200 Thu 12 Mar 2026 *in-person*] 25%
  - Project [presentation in week 13; report due Sun 19 Apr 2026] 25%
  - No final exam for this module!

Course Logistics

- Core Contents
  - Course materials and announcements on Canvas
  - Discussions on Canvas.
- Tutorials and Assignments
  - ~10 Tutorial Sessions:
    - Attempt tutorial questions before class; presentations and discussions in class
    - Submit group tutorial solutions after tutorial sessions
  - 2 Team Assignments: Written questions + Programming – Prerequisite: Python
- Project
  - Project Guidelines to be issued in Week 3
  - Thematic or Self-defined topic in teams of 3-5
  - 1 page proposal due after Recess Week

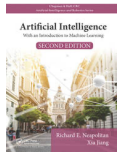
Reference Books



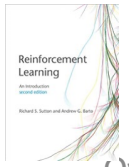


- Main reference book:
  - (RN) Russell, S. and P. Norvig, Artificial intelligence: A modern approach. 4th ed. (Global ed.) 2021: Pearson. [Kindle Edition] (Alternate: 3rd ed.)
    [Table of contents for 4th ed.: http://aima.cs.berkeley.edu/contents.html]
- Reference books:

  - (M) Murphy, K. P. (2023). Probabilistic machine learning: Advanced topics, MIT Press.[Book website: http://probml.github.io/book2]



  - (NJ) Neapolitan, R.E. and X. Jiang, Artificial intelligence : with an introduction to machine learning. Second edition. ed. Chapman & Hall/CRC artificial intelligence and robotics series. 2018, Boca Raton: CRC Press, Taylor & Francis Group. xiii, 466 pages.
    [Book website: https://doi.org/10.1201/b22400]



  - (SB) Sutton, R. S. and A. G. Barto. Reinforcement Learning: An introduction. 2nd ed. MIT Press, 2018, 2020
    [Book website: http://incompleteideas.net/book/the-book.html]
    [e-Book for personal use: http://incompleteideas.net/book/RLbook2020.pdf]

Additional Resources

- You will also find good tutorials, tools, publications at:
  - Conference in Uncertainty in Artificial Intelligence (UAI)
    - https://www.auai.org
  - American Association for Artificial Intelligence Conference (AAAI)
    - https://www.aaai.org
  - International Joint Conference on Artificial Intelligence (IJCAI)
    - https://www.ijcai.org
  - Neural Information Processing Systems Conference (NeurIPS)
    - https://nips.cc
  - International Conference on Automated Planning and Scheduling (ICAPS)
    - https://www.icaps-conference.org
  - International Conference on Autonomous Agents and Multiagent Systems (AAMAS)
    - https://www.ifaamas.org
  - International Conference on Artificial Intelligence and Statistics (AISTATS)
    - https://aistats.org
  - …

Honour Code

- NUS Code of Student Conduct:
  - (A) Academic, Professional and Personal Integrity
  - (B) Respect for People
  - (C) Respect for and Compliance with the Law and with Campus Policies and Regulations
  - (D) Responsibility towards Maintaining the Campus as a Place Conducive for Learning and Living
- This module will teach you how to apply and develop powerful Responsible AI technologies for the betterment of humankind
- If you are unable or unwilling to respect and abide by the Honour code, please **DO NOT** take this module!

NUS Course Materials:
Ethical Behaviour and Respecting Copyright

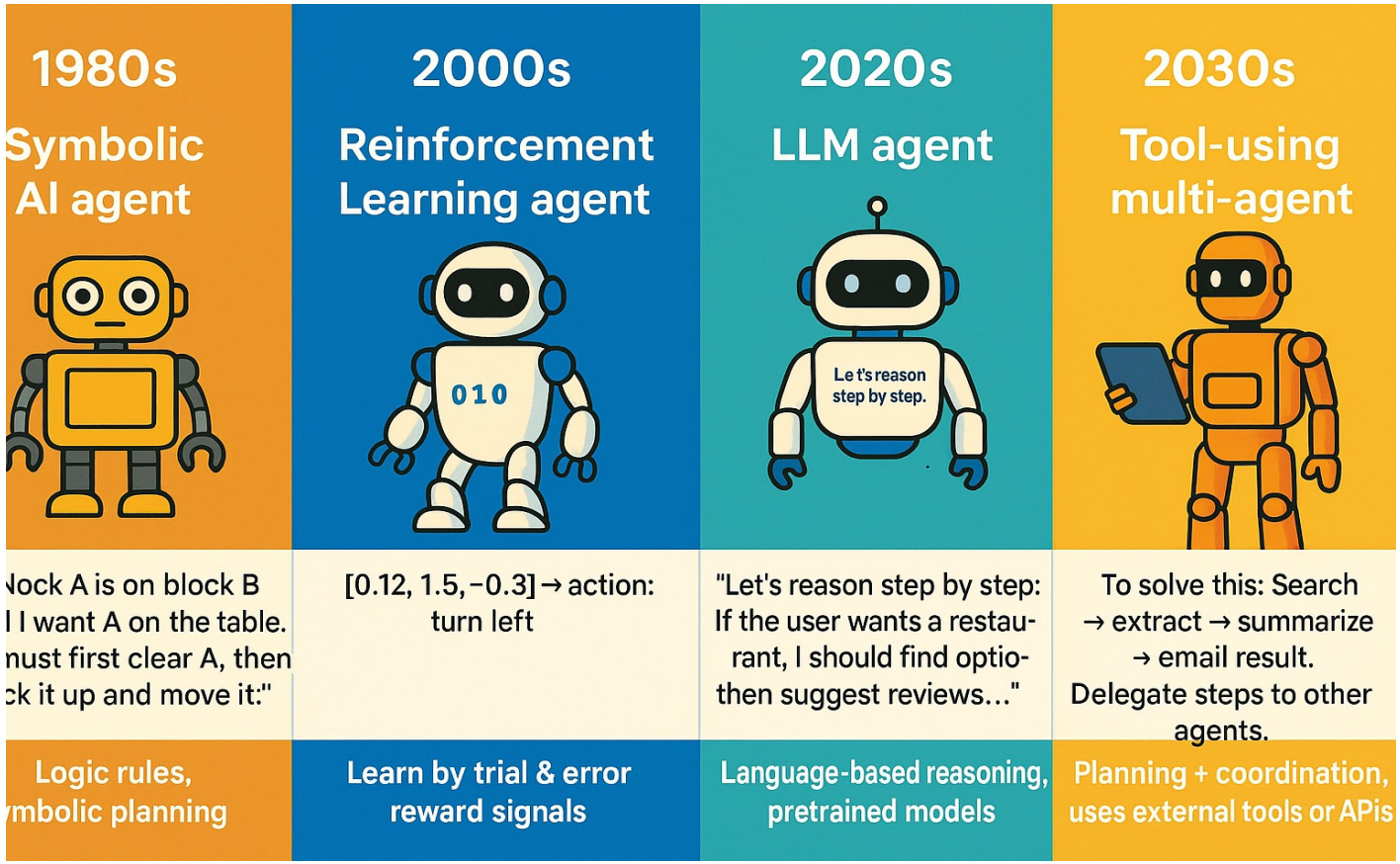
All course participants (including permitted guest students) who have access to the course materials on LumiNUS or any approved platforms by NUS for delivery of NUS modules are not allowed to re-distribute the contents in any forms to third parties without the explicit consent from the module instructors or authorized NUS officials

Examples of Disallowed Actions

- No Posting on any websites (except for the materials explicitly allowed by your lecturer in the respective module)
- No selling of material
- No sharing of questions/answers which could lead to cheating/plagiarism

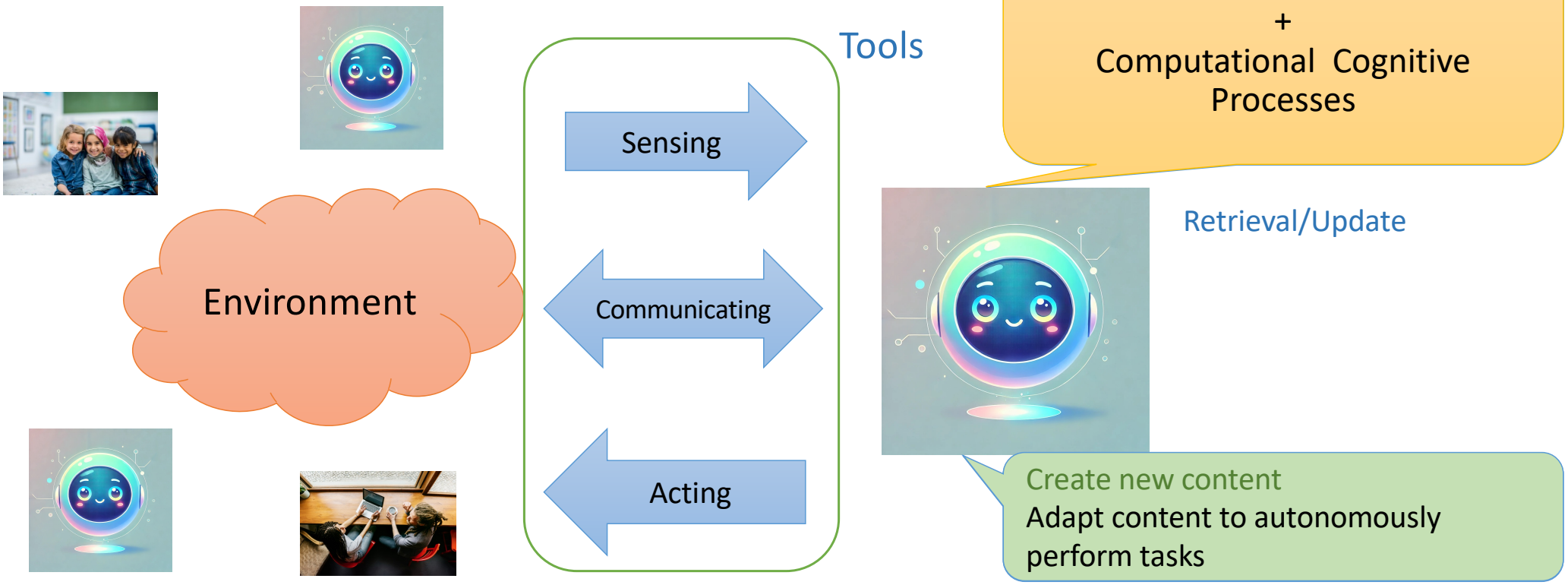
So, What Will I Learn in this Course?

A Brief History of Agents



Graphics generated by GPT4o

Agentic AI Landscape



Agent graphics source: ChatGPT4o, accessed 8 Aug 2024

Ultimate Goal: Multi-Agent Collaboration

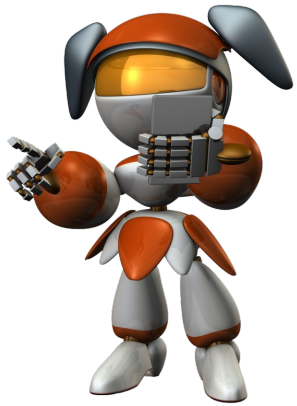
Multi-agent communication and coordination

Source: https://youtu.be/kopoLzvh5jY?si=7QYbkZcqzlw154gN

Beyond Technical Knowledge

- Domain challenges
  - Involving deep domain knowledge and operational issues
  - Interacting conditions, processes, and goals
- User challenges
  - Different skill levels and preferences
  - Varying usage patterns and cognitive biases
- Economic challenges
  - High implementation costs
  - Unclear market viability and scalability
- System challenges
  - Uncertain and changing information, processes, environments
  - Evolving IT and communication systems

A Grand Challenge!



How to design an AI agent
that can “grow” and be trained to play different roles?

- Educating RITA:
  - A **R**obust, **I**nteractive, **T**rustworthy, and **A**daptive (AI) Agent!

An early glimpse:
https://youtu.be/7pBOQeDrD5s?si=oofRtfwQih23Vibu